



BENEATH THE ICEBERG

The 80% AI Can't Build For You

Problem Statement 09 — THE NETWORK HUNTER

Hackathon Working Analysis • Judgment • Architecture • MVP • Demo Strategy

Core thesis

AI can dramatically accelerate the visible 20% of a hackathon build—UI, APIs, schemas, graph code, tests and documentation. For Network Hunter, the competitive advantage lives in the hidden 80%: defining the investigative problem precisely, modeling evidence and uncertainty, handling entity ambiguity, designing a time-aware workflow, choosing the right MVP, validating with realistic cases, and making the system defensible to judges.

Dimension	Network Hunter decision
Primary user	Investigator / intelligence analyst reviewing fragmented authorized records
Core pain	Relationships and patterns are difficult to discover across disconnected records
Recommended product	Evidence-first, time-aware investigation workspace
Differentiating workflow	Investigation Replay + source/evidence drill-down + human review
Safe prototype data	Synthetic investigative records, clearly labeled for demonstration
Positioning line	Connect. Replay. Verify.

Prepared as a practical build-order document for a student hackathon team. This document intentionally avoids claims that a prototype can determine guilt, predict individual criminality, or replace investigator judgment.

1. Executive Summary

Problem Statement 09 describes a classic fragmented-information problem: investigative records may contain people, organizations, transactions, communications, vehicles, locations and events, but the relationships between those records are hard to understand manually. The natural technical answer is a graph. The strategic answer is not simply “build a graph.”

The recommended concept is an evidence-first investigation platform that combines a relationship graph, chronological investigation replay, source-linked evidence, ambiguity handling and explainable pattern assistance. The graph becomes the navigation layer; the evidence and timeline become the trust layer.

The strongest hackathon narrative is therefore: fragmented records → connected entities → evolving network → inspect relationship → verify source → human review.

Assessment	Finding	Implication
Clarity	High	The PS is understandable, but the team must not drift into generic crime prediction.
Innovation	Moderate → strong if sharpened	Graph visualization is established; differentiate through evidence + time + workflow.
Impact	High	Faster review and better traceability are credible prototype outcomes.
Feasibility	High for prototype	Synthetic data can demonstrate the workflow without requiring operational police access.
AI fit	Strong as an assistant layer	Use AI for grounded summarization, search and pattern explanation—not autonomous judgment.
Main risk	Looking like “another graph tool”	Make Investigation Replay and evidence drill-down central to the demo.

2. The 20/80 Split Applied to Network Hunter

The 20% is what AI can accelerate. The 80% is what the team must decide, validate and defend.

AI-accelerated 20%	Human-owned 80%
UI screens and components	Which investigative workflow matters
Backend CRUD/API scaffolding	What each entity and relationship means
Graph rendering/query code	Observed vs inferred relationship semantics
CSV/JSON ingestion	Entity identity and ambiguity policy
NLP/search scaffolding	Evidence trust and provenance model
Test generation	Which real-world edge cases matter
Documentation/pitch drafts	Scope, trade-offs, freeze decisions and judge defense

Team rule: do not let an AI-generated feature list become your product roadmap. First decide the investigative job, then use AI to accelerate implementation.

The five most useful AI moves for this PS are: generate the interface; scaffold backend/data pipelines; generate graph queries and analytics; debug and test; and produce source-grounded summaries. The five most important human moves are: define the problem; define evidence semantics; define ambiguity handling; choose the MVP; and validate the result.

3. Before Coding: Problem Discovery, Validation & Empathy

Literal PS interpretation

The PS is asking for help with discovering hidden relationships and suspicious patterns from fragmented investigative information. It is not explicitly asking for a chatbot, a predictive-policing engine, facial recognition, or an autonomous suspect-ranking system.

Human validation exercise

Question	Team answer to lock
Who is the user?	Investigator / analyst who needs to connect and review records
What is fragmented?	People, organizations, accounts, communications, vehicles, locations, cases and events
What is hard manually?	Finding relationships, tracing them across time, and verifying their sources
What should the prototype improve?	Speed and clarity of relationship exploration and evidence review
What must remain human?	Interpretation, validation and investigative decisions

If domain access is available, interview at least one real investigator, analyst, retired practitioner, legal/domain expert or relevant institutional stakeholder. Ask what happens today, how records are connected, how identities are verified, and what makes a relationship trustworthy. Do not invent user findings from generic AI personas.

Key research insight to test

The team should test the hypothesis that the difficult part is not merely seeing connections, but understanding why a connection exists, when it existed, and which source record supports it.

4. Requirements & Scoping

Recommended MVP

Priority	Feature	Why it matters
☐ Must	Case dashboard	Defines the investigative context and avoids a generic graph demo.
☐ Must	Entity search	Lets the user start from a person, account, phone, vehicle, location or case.
☐ Must	Relationship graph	Core visual representation of interconnected records.
☐ Must	Timeline / Investigation Replay	Shows how the network evolves rather than presenting a static snapshot.
☐ Must	Evidence drawer	Connects every important relationship to source, date and record metadata.
☐ Must	Case/entity filtering	Controls graph clutter and focuses the investigation.
☐ Must	Ambiguity flags	Prevents name similarity from becoming false identity certainty.
☐ Should	Pattern assistant	Explains detected motifs/changes in plain language.
☐ Should	Natural-language search	Useful if grounded against structured data.
☐ Should	Report generation	Useful only after evidence traceability works.
☐ Avoid	Predictive crime / guilt scoring	High-risk, hard to validate and not necessary for the core PS.
☐ Avoid	Live CDR / national-scale integration	Requires authorization, security and infrastructure far beyond a hackathon MVP.
☐ Avoid	Blockchain / AR / facial recognition	Adds complexity without solving the central fragmented-record problem.

Scope test: Can we confidently demonstrate it live, explain how it works, explain its limitations, and recover if it fails? If not, it should not be a must-have.

5. System Design & Architecture

Recommended architecture



The key architectural principle is provenance: analytical views should preserve the path back to the underlying record. A relationship is not just an edge; it is an edge with type, time, source and status.

Suggested core entities

Entity	Suggested fields	Example role
Person	person_id, name, aliases, attributes	Human subject / contact
Organization	org_id, name, type	Company, group, institution
Account	account_id, type, institution	Financial or digital account
Vehicle	vehicle_id, registration	Movement/ownership context
Location	location_id, coordinates/address	Meeting or event context
Case	case_id, date, status	Investigative container
Event	event_id, timestamp, type	Call, transaction, meeting, movement
Source	source_id, type, date, reliability metadata	Record supporting an observation
Relationship	from_id, to_id, type, time, source_id, status	Observed/inferred link

Hard constraint

Never collapse two entities into one solely because their names are similar. The prototype should be able to say “possible match / ambiguous” and preserve human review.

6. Evidence Model: The Trust Layer

A graph-only product can show that A and B are connected. An evidence-first product shows what supports that connection.

Field	Purpose
relationship_id	Stable identifier for the relationship record
relationship_type	Called, transacted_with, associated_with, located_at, owned_by, etc.
from_entity / to_entity	The entities involved
observed_at / valid_from / valid_to	Temporal context
source_id	Pointer to the underlying record
source_type	CDR-like record, case record, transaction, vehicle log, etc.
verification_status	Observed / corroborated / inferred / ambiguous
notes	Human-review metadata

Example

Person A → Person B

Relationship: communication

Observed: 12 Mar 2026, 18:42

Source: CDR-1042

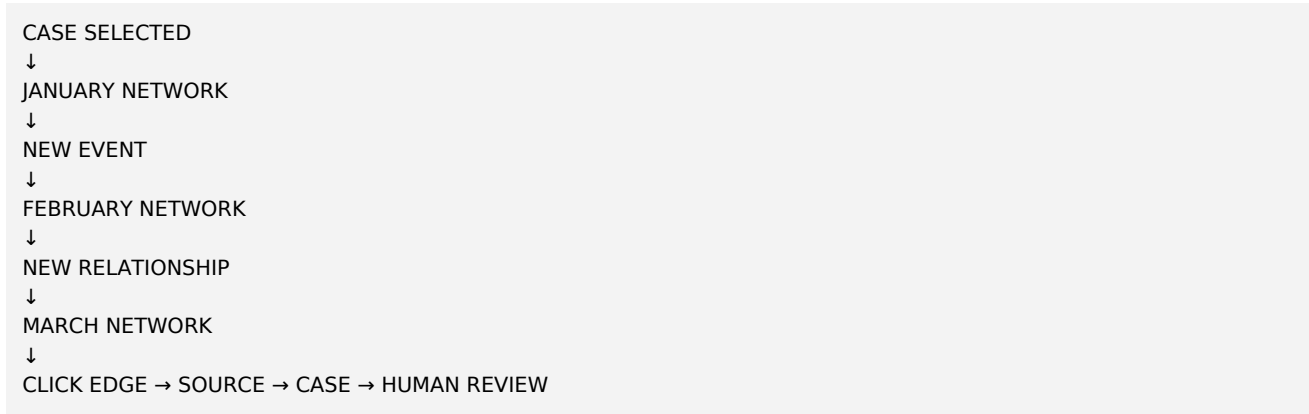
Status: Observed

Related case: CASE-019

This lets the interface answer six judge-friendly questions: WHO? WHAT? WHEN? WHY is it shown? WHAT SOURCE supports it? WHICH CASES are related?

7. Core Differentiator: Investigation Replay

The recommended innovation direction is to make the network time-aware and replayable. Instead of one static graph, the investigator can move through a date range and see which entities and relationships appeared, disappeared or changed.



Why this is strategically useful

Static graph problem	Replay response
Too many relationships at once	Filter by time and case
No sense of sequence	Show chronological events
Hard to explain why an edge matters	Open source evidence at the point in time
AI summary may feel opaque	Ground summaries in visible timeline records

Demo moment

Start with a small network. Move the timeline. A new relationship appears. Click it. The evidence drawer opens. Ask the AI assistant: “What changed between January and March?” The answer should enumerate changes and cite the underlying record identifiers—not make a conclusion about guilt.

8. Where AI Actually Belongs

AI should be an assistant to structured investigation, not the source of truth.

AI use	Implementation idea	Guardrail
Natural-language query	Translate a user question into filters/graph queries	Show the resulting records; do not hide retrieval.
Pattern explanation	Summarize graph motifs or temporal changes	Explain which rules/records produced the result.
Investigation summary	Generate a concise case/network summary	Ground only in retrieved records.
Relationship explanation	Answer “why are these entities connected?”	Return source IDs and dates.
Report drafting	Turn reviewed findings into structured notes	Mark generated text as draft for human review.

Avoid the phrase “AI detects criminals.” A defensible phrase is: “AI assists investigators by summarizing and explaining patterns already present in authorized records, with source references and human review.”

9. Data Availability & Synthetic Dataset Strategy

For a hackathon prototype, do not assume access to operational CDRs, confidential FIR/case files, criminal-history databases or restricted government systems. Build the workflow around synthetic investigative records unless the organizers explicitly provide authorized data.

Synthetic dataset design

Dataset	Suggested scale	Purpose
People	40-100	Create repeated names, aliases and legitimate ambiguity.
Organizations	10-25	Model affiliations and transactions.
Accounts	20-50	Create financial relationships.
Vehicles	15-30	Add movement/ownership links.
Locations	20-40	Add spatial/meeting context.
Cases	10-20	Provide investigative containers.
Events	100-300	Create temporal evolution.
Sources	100-300	Make relationships evidence-linked.

Synthetic-data label to display in the UI: “Synthetic Data — For Demonstration Only.”

The synthetic data should intentionally contain useful challenges: duplicate names, multiple identifiers per person, old relationships, contradictory records, missing values, relationships shared across cases, and a few planted network patterns that the system can explain.

10. Team Gamestorming Session

Run these games before implementation. The goal is to discover the smallest defensible product rather than the largest feature list.

Game 1 — Kill the Graph

For 10 minutes, forbid the word “graph.” Ask: “How else could we help an investigator understand connected evidence?” Candidate answers: timeline, evidence explorer, investigation replay, relationship search, case story builder, investigation copilot. Then bring the graph back as the navigation layer.

Game 2 — 5 Whys

Why do investigators need network analysis? → Because information is fragmented. Why does fragmentation hurt? → Relationships are hard to see. Why does that matter? → Important patterns can be missed or take longer to investigate. Why? → Humans must manually cross-reference many records. Why? → The information lacks a unified, traceable investigation view.

Game 3 — Persona Switch

Persona	Question
Investigator	Can I understand what connects these entities?
Analyst	Can I discover patterns across cases and time?
Reviewer	Can I verify where this relationship came from?
Supervisor	Can I understand what the team has reviewed and why?

Game 4 — 3-Click Test

Select Case → Select Entity → Select Relationship. Within three clicks, the system should expose WHO / WHAT / WHEN / SOURCE / RELATED CASE. If it cannot, the UX is probably too complex.

11. Edge Cases & Judge Attack

Judge attack	Expected design response
Two people have the same name	Do not merge by name alone; flag ambiguity and show supporting identifiers.
One person has multiple phone numbers	Model identifiers as connected records rather than separate people.
Relationship is years old	Use timestamps/date ranges and timeline filtering.
Two records contradict each other	Preserve both sources and show conflict rather than silently choosing one.
AI explanation is wrong	Ground generation in retrieved records and source IDs.
Dataset becomes too large	Use case/time filters and progressive graph expansion.
Judge asks whether a link proves guilt	State clearly that a relationship is an analytical observation, not proof of guilt.
Judge asks for real police data	Explain that operational data requires authorization; prototype uses synthetic data.
Judge asks what is innovative	Demonstrate Investigation Replay + evidence drill-down + ambiguity handling.

Reality-check testing

Have one teammate who did not build a feature perform the complete workflow from a fresh start. Test missing fields, duplicate names, contradictory sources, empty search results, large graphs, incorrect dates and AI questions that cannot be answered from the available evidence.

12. Competitive Positioning

Criminal-network visualization and link analysis are established categories. Products such as IBM i2 Analyst's Notebook and Maltego demonstrate that graph/link analysis itself is not a new invention. Therefore, the pitch should not claim "we invented criminal network graphs."

Category	What is established	Network Hunter opportunity
Link analysis	Nodes and relationships for investigative analysis	Make the relationship directly traceable to evidence.
OSINT/entity analysis	Entity/link exploration and data transformations	Focus the prototype on a coherent case workflow.
Graph infrastructure	Graph storage, querying and analytics	Use graph technology as infrastructure, not as the entire product.
AI assistants	Natural-language summarization/search	Ground every answer in visible structured records.
Timeline analysis	Temporal views exist in investigative tools	Make timeline + graph replay the central interaction.

Positioning statement

"We are not claiming criminal network visualization is new. Network Hunter focuses on an evidence-first investigative workflow that combines relationship exploration, timeline replay, source-linked records and human review in one simplified prototype."

13. Recommended Build Order

Stage	Build	Definition of done
1	Data schema + seed dataset	Synthetic records load consistently.
2	Backend/API	Cases, entities, relationships, sources retrievable.
3	Graph view	Entity search opens connected network.
4	Evidence drawer	Every selected edge can reveal source/date/status.
5	Timeline	Network/events can be filtered or replayed by time.
6	Ambiguity handling	Duplicate/uncertain entities are visibly flagged.
7	Pattern engine	At least 2-3 explainable patterns work.
8	AI layer	Answers grounded questions using retrieved records.
9	Demo hardening	Fresh-user test, failure cases, backup dataset/video.
10	Pitch	2-minute story matches the actual working flow.

Recommended demo sequence

1) Open a case → 2) search an entity → 3) show network → 4) move the timeline → 5) reveal a new relationship → 6) click it → 7) show source evidence → 8) ask a grounded question → 9) show human-review state → 10) summarize the investigation workflow.

14. Team Alignment & Ownership

Create one owner per deliverable and one person prepared for every judge-question category. Do not rely on “everyone knows everything.”

Area	Primary owner	Backup / review
Problem statement + domain	Team lead / domain owner	All members
Data model + synthetic data	Backend/data owner	AI/ML owner
Graph + analytics	Backend/ML owner	Frontend owner
Frontend + UX	Frontend owner	Demo owner
AI assistant	AI/ML owner	Backend owner
Testing	Dedicated tester	All members
Pitch + demo script	Pitch/demo owner	Team lead
Architecture Q&A;	Backend/architecture owner	AI/ML owner
Data/privacy Q&A;	Data/domain owner	Team lead

Before submission, every member should independently answer: What problem are we solving? What is innovative? What data are we using? How does the graph avoid false identity matches? How does AI get its evidence? What happens when the system is wrong? Why can't an existing graph tool do this? What is our human-review boundary?

15. High-Pressure Hackathon Schedule

Window	Focus	Output
0:00-0:30	Problem lock	One-sentence problem, user, success metric, hard constraints.
0:30-1:15	Data/model	Schema + synthetic records + planted scenarios.
1:15-2:45	Core backend/graph	Working APIs + graph exploration.
2:45-3:45	Evidence + timeline	Source drawer + time filtering/replay.
3:45-4:30	Pattern layer	2-3 explainable patterns.
4:30-5:15	AI assistant	Grounded summary/query workflow.
5:15-6:00	Integration	Full end-to-end demo path.
6:00-6:30	Break/failure testing	Fresh-user test + failure cases.
6:30-7:00	Freeze + pitch	No new features; rehearse and backup.

If your hackathon lasts longer, use the extra time for validation, UX, test coverage and demo reliability before adding new architecture.

16. Judge Q&A; Stress Test

Question	Answer structure
What problem are you solving?	Investigators face fragmented records; Network Hunter connects them into a time-aware, evidence-linked view.
What is innovative?	Not the graph itself; the integrated evidence-first Investigation Replay workflow and explicit human review.
Where is AI?	Grounded natural-language search, pattern explanation and summaries over structured records.
Why not just use a graph database?	A graph database is infrastructure; our prototype adds the investigator workflow, timeline, evidence and review layer.
Why not just use Maltego/i2?	We are not claiming replacement. We demonstrate a focused workflow around source-linked replay and simplified case review.
Do you have real police data?	Operational data requires authorization; the prototype uses synthetic records to safely demonstrate the analytical workflow.
Can this prove someone is a criminal?	No. A relationship or pattern is not proof of guilt; the system supports investigation and human review.
What if AI hallucinates?	Answers are grounded in retrieved records and source IDs; unsupported claims should be rejected.
How does it scale?	Separate ingestion, graph/query and UI layers; production scale would require authorized infrastructure, security and governance.

17. Final Go / No-Go Checklist

Gate	Pass condition	Target
Clarity	Every member gives the same 30-second explanation	☐
User validation	At least one credible domain insight or explicitly labeled hypothesis	☐ ☐
Innovation	Team can explain differentiation without claiming graph analysis is new	☐
Data	Synthetic dataset or authorized dataset is ready and clearly labeled	☐
Evidence	Important relationships have source/date/status metadata	☐
Temporal workflow	Timeline/replay works on the main demo case	☐
AI grounding	AI cannot freely invent investigative facts	☐
Edge cases	Duplicate identity, missing data, contradiction and empty results tested	☐
Demo reliability	Backup dataset/video/screens available	☐
Scope freeze	No uncontrolled features in the final hours	☐
Judge defense	Every member knows data, architecture, innovation and limitation answers	☐

Decision rule

Proceed with Network Hunter if the team can demonstrate the complete evidence-linked workflow reliably. Do not lock the concept merely because the graph looks impressive. Lock it when the problem, user, data model, differentiation, MVP and demo story all agree.

18. The Network Hunter Iceberg



The strategic lesson

For Network Hunter, the visible 20% is easy to recognize: a graph, a timeline and an AI assistant. The hidden 80% is what makes those components meaningful: knowing what a relationship means, knowing what evidence supports it, knowing when it existed, knowing when the system is uncertain, and knowing what the human investigator must decide.

Final positioning: “Network Hunter transforms fragmented investigative records into a time-aware, evidence-linked network that investigators can explore, replay and review.”

Tagline: Connect. Replay. Verify.

End of working analysis.